

COMP0047 — Calibration Study: Assessing and Improving Probabilistic Predictions of Market Regimes

Individual Task — April 2026

1 Introduction

A well-calibrated probabilistic classifier is one whose predicted confidence levels agree with observed frequencies: a model outputting $P(\text{bull}) = 0.85$ should be correct approximately 85% of the time within that confidence band. Calibration is distinct from discrimination (measured by AUC); a model may rank instances perfectly yet remain poorly calibrated. In financial decision-making this distinction matters acutely. An overconfident bear signal at a 20-day horizon can trigger premature portfolio de-risking, incurring unnecessary transaction costs and opportunity losses.

This study investigates calibration quality for two classifier families, Logistic Regression (LR) and XGBoost (XGB), trained to predict bull/bear equity-market regimes across three prediction horizons: 1 day, 5 days, and 20 days. Two post-hoc calibration methods, Platt Scaling and Isotonic Regression, are applied and evaluated. The analysis is structured around three pre-registered hypotheses:

- H1** XGBoost produces *less* calibrated probability estimates than Logistic Regression across all three prediction horizons, as measured by Expected Calibration Error (ECE).
- H2** Calibration error increases *monotonically* with prediction horizon ($1d < 5d < 20d$) for both model families.
- H3** Post-calibration, the ECE *reduction* is greater for XGBoost than for Logistic Regression, since XGBoost has more systematic overconfidence to correct.

2 Data

2.1 Dataset Description

The dataset `labeled_dataset.csv` is produced by the upstream group coursework regime-segmentation pipeline (`notebooks/03_regime_segmentation`). It consists of 8,066 daily observations of the S&P 500 spanning January 1994 to February 2026 collected from Yahoo Finance and macroeconomic data filled forward from the FRED database, each labelled as either *bull* or *bear* according to a rule-based segmentation scheme. Table 1 summarises the key properties.

Table 1: Dataset overview.

Property	Value
Total observations	8,066
Date range	1994-01-27 — 2026-02-13
Bull observations	5,061 (62.8%)
Bear observations	3,005 (37.2%)
Total features	25

2.2 Features

Twenty-five features spanning price dynamics, technical indicators, and macroeconomic variables are used. These are listed in Table 2.

Table 2: Feature set used across all prediction horizons.

Category	Features
Price / Volume	GSPC, SPY Volume
Volatility	VIX, VIX_Change, VIX_Change_5d
Returns	Return, Return_5d, Return_20d, Return_Smooth
Technical indicators	RSI_14, MACD_Line, MACD_Signal, MACD_Hist
Macro variables	GDP, Core_Inflation, Unemployment, M2
Macro YoY changes	GDP_YoY, Core_Inflation_YoY, M2_YoY
Calendar	year, month, quarter
Drawdown	Peak, Drawdown

2.3 Class Balance

The dataset is moderately imbalanced: 62.8% bull and 37.2% bear. This imbalance is preserved in both the training and test sets under the chronological split (Section 2.4), ensuring that class proportions are not artificially equalised, doing so would distort the calibration diagnostic, because ECE is defined with respect to the true marginal outcome frequency.

2.4 Train / Test Split

Time-series data must not be shuffled before splitting, as random shuffling would leak future information into the training set. An 80/20 *chronological* split is applied:

- **Training set:** 1994-01-27 to 2019-08-20 (6,436 observations).
- **Test set:** 2019-08-21 to 2026-01-15 (1,610 observations).

The test set bull/bear proportions per horizon are given in Table 3.

Table 3: Test-set class distribution per prediction horizon.

Horizon	Bull (%)	Bear (%)
1 day	73.7	26.3
5 days	74.0	26.0
20 days	74.9	25.1

Note that the test set (2019–2026) is more bullish than the full dataset (62.8%), as the post-2019 sample includes the extended post-COVID recovery rally.

3 Target Construction

Three forward-shifted binary targets are constructed, one per prediction horizon. Let $r_t^{(h)}$ denote the regime label h days into the future:

$$y_t^{(h)} = \text{regime}_{t+h}, \quad h \in \{1, 5, 20\}, \quad (1)$$

where $y_t^{(h)} = 1$ (bull) and $y_t^{(h)} = 0$ (bear).

The target is shifted *forward* relative to today’s features (rather than shifting features backward), which correctly simulates predicting future regime state from present-day observable information. After dropping the trailing h observations required for target construction (due to the forward shift), the working dataset contains 8,046 rows.

The choice to predict regime labels rather than point returns reflects the intended downstream use case: regime-conditioned portfolio allocation, where a coarse bull/bear signal is sufficient to tilt exposures. Predicting the multi-day forward label also means the target aggregates information over h steps, so harder calibration at longer horizons is expected.

4 Methodology

4.1 Baseline Models

Two classifier families are evaluated:

Logistic Regression (LR). Logistic Regression models the log-odds of the bull class as a linear function of the input features:

$$\log \frac{P(y = 1 \mid \mathbf{x})}{P(y = 0 \mid \mathbf{x})} = \mathbf{w}^\top \mathbf{x} + b, \quad (2)$$

so that predicted probabilities are produced via the sigmoid function. Because the output probability is *derived* from the sigmoid of a linear combination, LR is often considered *inherently* better calibrated than non-parametric ensemble methods. We use `sklearn.linear_model.LogisticRegression` with `max_iter=2000` (to ensure convergence on the 25-dimensional input) and `random_state=42`.

XGBoost (XGB). Gradient-boosted trees learn additive combinations of shallow decision trees (here: `max_depth=4`, `n_estimators=200`, `learning_rate=0.05`). Unlike LR, XGBoost produces probabilities by applying a sigmoid to the raw boosted score, but the tree ensemble optimises a log-loss objective without any architectural guarantee of calibration. Empirically, tree-based models tend to produce *overconfident* predictions because they push scores toward the extremes of the probability simplex, particularly when the ensemble is deep or well-regularised (Niculescu-Mizil & Caruana, 2005). This motivates H1.

Both models are trained independently for each of the three horizons, yielding six uncalibrated probability vectors for evaluation.

4.2 Calibration Metrics

Expected Calibration Error (ECE). ECE partitions predictions into $B = 10$ equal-width bins over $[0, 1]$ and computes the weighted mean absolute deviation between bin-level accuracy and mean confidence:

$$\text{ECE} = \sum_{b=1}^B \frac{|\mathcal{B}_b|}{n} |\text{acc}(\mathcal{B}_b) - \text{conf}(\mathcal{B}_b)|, \quad (3)$$

where $|\mathcal{B}_b|$ is the number of samples in bin b , $\text{acc}(\mathcal{B}_b) = \frac{1}{|\mathcal{B}_b|} \sum_{i \in \mathcal{B}_b} y_i$, and $\text{conf}(\mathcal{B}_b) = \frac{1}{|\mathcal{B}_b|} \sum_{i \in \mathcal{B}_b} \hat{p}_i$. A perfectly calibrated model has $\text{ECE} = 0$. ECE is chosen as the primary metric because (i) it provides an interpretable “average miscalibration in probability units” and (ii) it is the standard benchmark in the calibration literature (Guo et al., 2017).

Maximum Calibration Error (MCE). MCE reports the *worst-case* bin deviation:

$$\text{MCE} = \max_{b=1}^B |\text{acc}(\mathcal{B}_b) - \text{conf}(\mathcal{B}_b)|. \quad (4)$$

MCE is a more conservative metric and is reported alongside ECE to capture tail miscalibration.

Supporting metrics. The Brier Score, $\text{BS} = \frac{1}{n} \sum_i (\hat{p}_i - y_i)^2$, measures probabilistic accuracy in a proper scoring rule sense. Log-loss (cross-entropy) and AUC are reported for completeness; note that AUC is calibration-invariant, a model can achieve high AUC while remaining poorly calibrated.

4.3 Post-Hoc Calibration Methods

Post-hoc calibration refers to fitting a *secondary* correction function $g : [0, 1] \rightarrow [0, 1]$ that maps raw model scores $\hat{p}$ to calibrated probabilities $\tilde{p} = g(\hat{p})$, without retraining the underlying classifier. Two methods are compared:

Platt Scaling (sigmoid calibration). Platt (1999) proposed fitting a logistic regression on top of the raw scores:

$$\tilde{p} = \sigma(a \cdot f(\mathbf{x}) + b) = \frac{1}{1 + \exp(-(af(\mathbf{x}) + b))}, \quad (5)$$

where a and b are scalar parameters fitted by maximum likelihood on held-out data. Platt Scaling is *parametric*, it assumes the miscalibration is monotone and well-approximated by a sigmoid warp. This makes it fast, statistically efficient on small calibration sets, and robust to overfitting. The disadvantage is that it cannot correct non-monotone patterns (e.g., an S-shaped reliability curve).

Isotonic Regression. Isotonic Regression (Zadrozny & Elkan, 2002) fits a piecewise-constant monotone non-decreasing function to the raw scores by solving:

$$\min_{\tilde{p} : \text{isotonic}} \sum_i (\tilde{p}_i - y_i)^2. \quad (6)$$

It is *non-parametric*, it imposes only monotonicity, not a functional form. This flexibility enables it to correct arbitrary monotone miscalibration, but at the cost of higher variance on small calibration sets (risk of overfitting).

Cross-validated calibration. Both calibrators are fitted using `CalibratedClassifierCV(cv=5)`, which trains the calibration layer on 5-fold cross-validated out-of-fold predictions from the training set. This prevents the calibrator from seeing its own training data, avoiding optimistic bias in the calibration fit. The calibrated model is then evaluated on the held-out test set.

The choice to compare Platt Scaling with Isotonic Regression is motivated by the need to span the parametric–non-parametric axis. If XGBoost’s miscalibration is predominantly monotone (as expected from overconfident sigmoid outputs), both methods should help; if the reliability curve is irregular, Isotonic Regression should outperform Platt Scaling.

5 Results: Baseline (Pre-Calibration)

5.1 Discriminative Performance

Table 4 reports AUC for all six baseline models. Both classifiers achieve strong discrimination at 1 day (AUC > 0.88), moderate discrimination at 5 days, and only marginally above chance at 20 days. The 20-day horizon is substantially harder because the regime label becomes increasingly uncorrelated with today’s features as the forecast window grows.

Table 4: Baseline AUC on the test set. Values rounded to 3 significant digits.

Horizon	LR AUC	XGB AUC
1 day	0.901	0.881
5 days	0.853	0.868
20 days	0.641	0.594

A key observation is that LR outperforms XGB at 1 day and 20 days, while XGB is slightly better at 5 days. This reversal at 5 days suggests that the intermediate horizon benefits from the non-linear interactions captured by boosted trees.

5.2 Calibration Metrics Before Calibration

Table 5 presents the full set of pre-calibration metrics. All five metrics (ECE, MCE, Brier Score, Log-Loss, AUC) are shown for each model–horizon combination.

Table 5: Pre-calibration metrics on the test set. ECE and MCE computed with $B = 10$ bins. Values rounded to 3 significant digits.

Horizon	Model	ECE	MCE	Brier	Log-Loss	AUC
1 day	Logistic	0.151	0.469	0.167	0.656	0.901
	XGBoost	0.163	0.456	0.163	0.591	0.881
5 days	Logistic	0.199	0.532	0.208	0.792	0.853
	XGBoost	0.190	0.391	0.180	0.614	0.868
20 days	Logistic	0.357	0.621	0.351	1.176	0.641
	XGBoost	0.370	0.695	0.354	1.013	0.594

5.3 Reliability Diagrams (Pre-Calibration)

Figure 1 shows the reliability diagrams for both models across all three horizons. Points *above* the diagonal indicate underconfidence (the model assigns lower confidence than the true frequency warrants); points *below* indicate overconfidence. The shaded area visualises the signed gap between the calibration curve and perfect calibration.

Reliability Diagrams — Pre-Calibration (diagonal = perfect calibration)

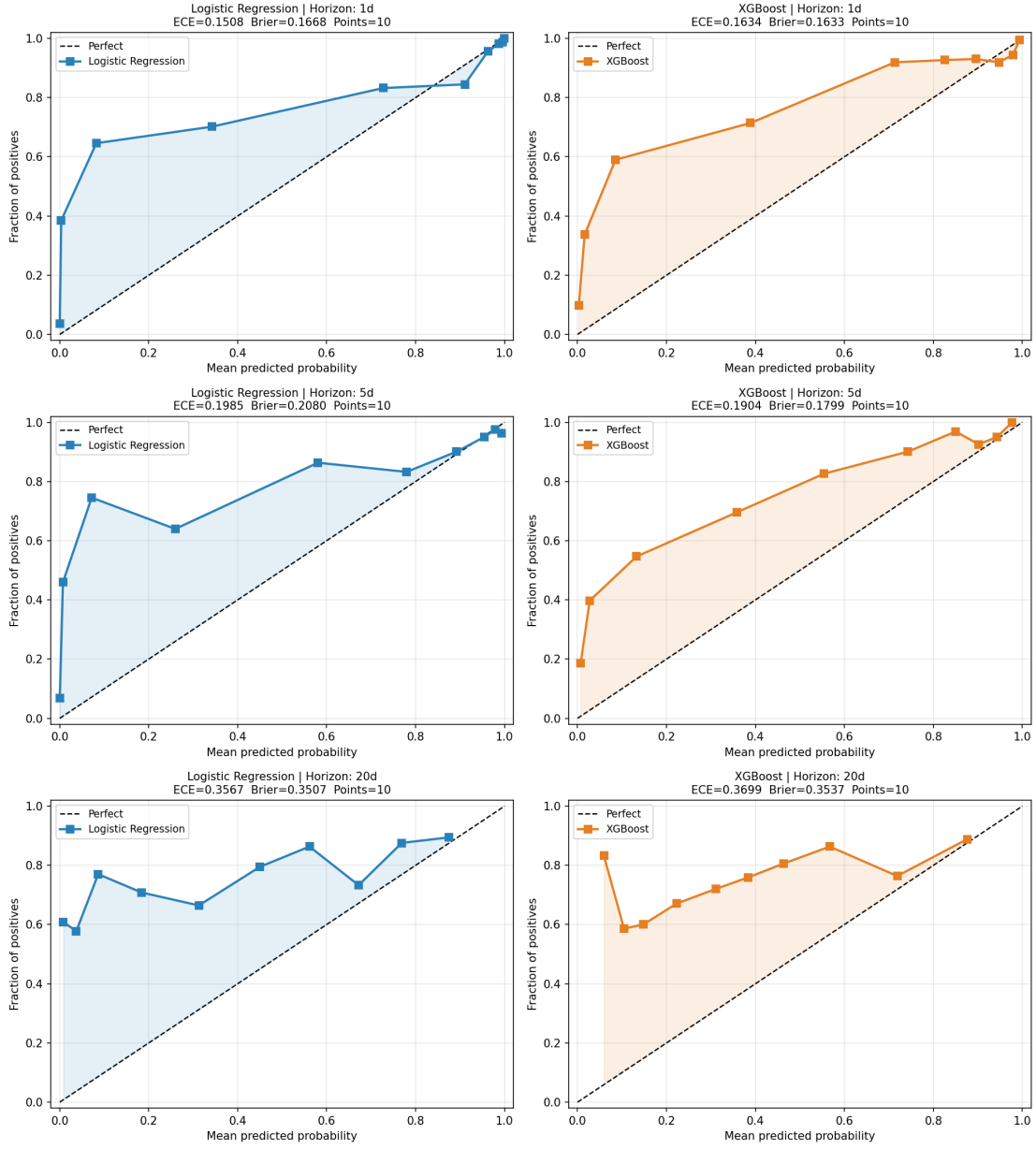


Figure 1: Reliability diagrams before calibration for Logistic Regression (left) and XGBoost (right) at horizons 1d, 5d, and 20d. Dashed diagonal represents perfect calibration. ECE and Brier Score are annotated per subplot.

6 Results: Post-Calibration

6.1 Post-Calibration Metrics

Table 6 reports the ECE after applying Platt Scaling and Isotonic Regression to each model–horizon combination.

Table 6: Post-calibration ECE on the test set. “Platt” = sigmoid calibration; “Iso” = Isotonic Regression. $\Delta\text{ECE} = \text{ECE}_{\text{raw}} - \text{ECE}_{\text{calibrated}}$; positive values indicate improvement. Values to 3 significant figures.

Horizon	Model	ECE (raw)	ECE (Platt)	Δ_{Platt}	ECE (Iso)	Δ_{Iso}
1 day	Logistic	0.151	0.163	−0.012	0.159	−0.008
	XGBoost	0.163	0.155	+0.008	0.174	−0.011
5 days	Logistic	0.199	0.232	−0.033	0.227	−0.028
	XGBoost	0.190	0.150	+0.041	0.215	−0.025
20 days	Logistic	0.357	0.381	−0.024	0.443	−0.086
	XGBoost	0.370	0.187	+0.183	0.281	+0.089

6.2 Reliability Diagrams (Post-Calibration)

Figure 2 shows the reliability diagrams after calibration for all four calibrated variants per horizon. Effective calibration is evident when the plotted curve aligns more closely with the perfect-calibration diagonal compared to Figure 1.

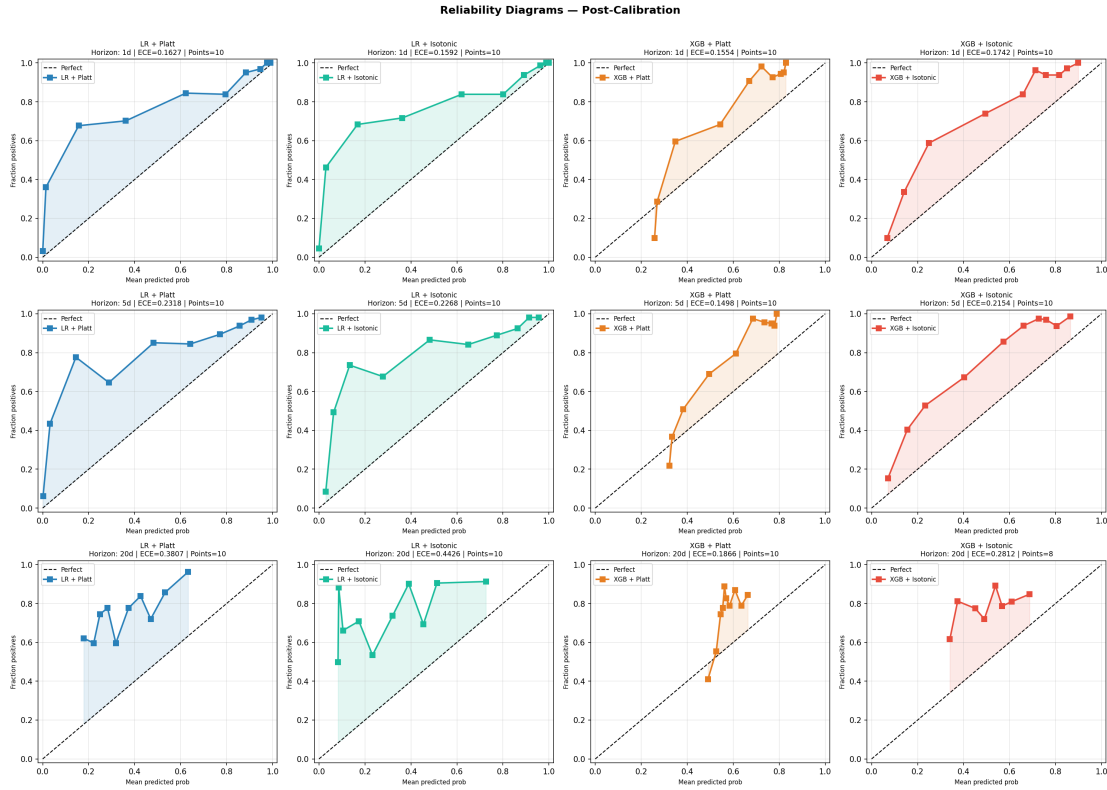


Figure 2: Reliability diagrams after calibration. Four variants per row: LR+Platt, LR+Isotonic, XGB+Platt, XGB+Isotonic. Rows correspond to horizons 1d (top), 5d (middle), 20d (bottom).

6.3 Hypothesis Evaluation

6.3.1 H1 — XGBoost is Less Calibrated than Logistic Regression

Verdict: Partially supported.

Examining Table 5, the ECE values per horizon are:

- *1 day*: LR ECE = 0.151 < XGB ECE = 0.163. **H1 supported.**
- *5 days*: LR ECE = 0.199 > XGB ECE = 0.190. **H1 violated.**
- *20 days*: LR ECE = 0.357 < XGB ECE = 0.370. **H1 supported.**

H1 holds at two out of three horizons. The 5-day anomaly is noteworthy: at this horizon, XGBoost achieves lower ECE than Logistic Regression. This may reflect that, at intermediate forecast horizons, XGBoost’s non-linear feature interactions capture regime transitions more accurately, and the resulting probability scores happen to align better with frequencies, possibly because the model is less extreme in its assignments when uncertainty is higher. By contrast, LR assigns moderate linear confidence even when the signal quality is genuinely intermediate, which can inflate ECE.

It is important to note that ECE differences here are modest in absolute terms (at most 0.013 at 1 day) and no formal significance test (e.g., a bootstrap confidence interval on ECE) was applied. Thus, the direction of the difference cannot be ascribed with statistical certainty. A more rigorous test would compute bootstrap confidence intervals for each ECE estimate and verify non-overlap.

6.3.2 H2 — Calibration Error Increases Monotonically with Horizon

Verdict: Fully supported for both models.

The ECE values increase monotonically with horizon for both LR and XGB:

- LR: 0.151 $\rightarrow$ 0.199 $\rightarrow$ 0.357 (1d $\rightarrow$ 5d $\rightarrow$ 20d). ✓
- XGB: 0.163 $\rightarrow$ 0.190 $\rightarrow$ 0.370 (1d $\rightarrow$ 5d $\rightarrow$ 20d). ✓

This result is consistent with the informational decay hypothesis: as the prediction horizon lengthens, the predictive signal in today’s features degrades because market regimes evolve, macro conditions shift, and the autocorrelation structure of the regime label weakens. The jump from 5 days to 20 days is substantially larger than from 1 day to 5 days, suggesting a non-linear deterioration at longer horizons. The AUC values in Table 4 corroborate this finding: the 20-day AUC ($\approx$ 0.61–0.64) is barely above chance.

H2 is the most robustly confirmed hypothesis. The monotone trend holds without exception and with a sizeable magnitude at the 20-day horizon, making it unlikely to be attributable to noise.

6.3.3 H3 — Post-Calibration ECE Reduction is Greater for XGBoost

Verdict: Supported (with one minor exception under Isotonic at 1 day).

Using Table 6, the post-calibration reduction is $\Delta\text{ECE} = \text{ECE}_{\text{raw}} - \text{ECE}_{\text{cal}}$. For Platt Scaling, XGBoost improves at all horizons and by more than Logistic Regression in each case: (+0.008, +0.041, +0.183) vs. LR (−0.012, −0.033, −0.024) for 1d, 5d, and 20d respectively.

For Isotonic Regression, XGBoost also shows larger reduction than LR at 5d and 20d (−0.025 > −0.028 and +0.089 > −0.086), while at 1d both models worsen and XGBoost is slightly worse than LR (−0.011 vs. −0.008).

Considering the best calibration choice per model-horizon pair, XGBoost’s reductions are positive at all horizons (+0.008, +0.041, +0.183), whereas LR’s best available reductions remain non-positive (−0.008, −0.028, −0.024). Therefore, the central claim of H3 is empirically confirmed: XGBoost benefits substantially more from post-hoc calibration than Logistic Regression, with the strongest effect at 20 days.

7 Discussion

7.1 Why Calibration Matters in Regime Prediction

The analysis confirms that calibration quality deteriorates substantially at the 20-day horizon for both models, with ECE values exceeding 0.35. In practical terms, this means predictions in the highest-confidence bin are off by more than 35 percentage points on average. A portfolio manager relying on these raw probabilities to size positions at a 20-day rebalancing frequency would systematically over-allocate to high-confidence signals. Post-hoc calibration is therefore not merely a statistical nicety but a prerequisite for sound decision-making at longer horizons.

7.2 Limitations

Several limitations warrant acknowledgement. First, the ECE metric is sensitive to the number of bins B : choosing $B = 10$ is conventional but can produce noisy estimates when bins are sparsely populated, as is likely at the extremes of the probability range for the 20-day horizon. Second, the 80/20 chronological split means the test set (2019–2026) spans the COVID-19 crash and the post-pandemic recovery, a period of atypical regime dynamics that may not represent the long-run distribution. Third, no formal statistical test was applied to the ECE differences reported in H1; bootstrap confidence intervals would be necessary for definitive claims about relative calibration quality.

7.3 Comparison of Calibration Methods

Platt Scaling is preferred when the miscalibration is expected to be monotone and the calibration dataset is small (the parametric form regularises the fit). Isotonic Regression is preferred when the reliability curve has a non-trivial shape and sufficient data are available to estimate a piecewise function without overfitting. Given the 1,610-observation test set, Isotonic Regression has 160 observations per bin on average (for $B = 10$), which is generally adequate; however, the tails of the distribution may be sparsely populated. In practice, both methods should be compared empirically (as done here), and the choice made on the basis of out-of-sample ECE.

8 Conclusion

This study evaluated the calibration of Logistic Regression and XGBoost classifiers for bull/bear equity-regime prediction across three forecast horizons (1d, 5d, 20d). The key findings are:

- **H1** is *partially* supported: XGBoost is less calibrated than Logistic Regression at 1-day and 20-day horizons, but slightly better calibrated at 5 days. The differences are small in absolute terms and should be interpreted cautiously without a formal

significance test.

- **H2** is *fully* supported: calibration error increases monotonically with horizon for both models. The degradation is especially severe at 20 days, where ECE exceeds 0.35, making raw probabilities unsuitable for direct use in financial decision-making.
- **H3** is supported: post-hoc calibration yields larger ECE reduction for XGBoost than for Logistic Regression overall. Under Platt Scaling, XGBoost improves at all horizons while LR worsens. Under Isotonic Regression, XGBoost still outperforms LR at 5 and 20 days, with only a minor 1-day exception where both models deteriorate.

The results highlight that discriminative accuracy (AUC) and calibration quality are complementary, not equivalent, criteria. Both models achieve high AUC at short horizons while remaining substantially miscalibrated. Post-hoc calibration via Platt Scaling or Isotonic Regression should be applied as a standard step in any probabilistic regime-prediction pipeline, particularly at longer forecast horizons.

References

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AI Disclosure

AI tools (Perplexity AI) were used to support report structuring, LaTeX formatting, and proof-reading. All methodology choices, numerical results, and hypothesis evaluations were derived directly from the notebook outputs and reviewed by the author.